IntelliSim — Simulation Analysis Report

exp-G-med-rep1

Model: exp-G-med-b0.18-model

Analysis language: en

LLM: openai · o4-mini

Simulation status at analysis: finished

Steps covered: 0–227 (sampling 1, 77 points)

Context: Analyze this simulation's results. Context: An infectious disease spreads in a population of moving agents. Describe the observed dynamics, quantify the key outcomes, and note anything notable or unexpected.

Executive Summary

The simulation produced a classic SIR epidemic curve in a fully mixed population of 200 agents on a toroidal grid. An initial exponential-like rise led to a peak of 22 simultaneous infections at step 60, driven by proximity encounters during random movement. Recovery began to outpace new infections as the susceptible pool declined from 198 to 165, causing a primary collapse to a single active case by step 138. Despite only 37% of agents ever recovering, transmission fell below a reproduction threshold, extinguishing the main wave.

After the primary decline, residual infections seeded two minor rebound waves between steps 147–180 and 198–207, peaking at 9 and 7 cases respectively. Uniform random motion (45 600 moves) prevented persistent spatial hotspots and kept all classes—susceptible, infected, recovered—thoroughly intermixed. By conclusion (step 227) no infected remained, with 74 recovered and 126 still susceptible.

Simulation & Model Overview

Spatial: true. Update mode: sequential. Archetypes: 3. Rules: 4. Interactions: 0. Events: 0. Environment: 50x50 (torus=true, shapes=0). Archetypes: Susceptible Person, Infected Person, Recovered Person.

Quantitative Metrics

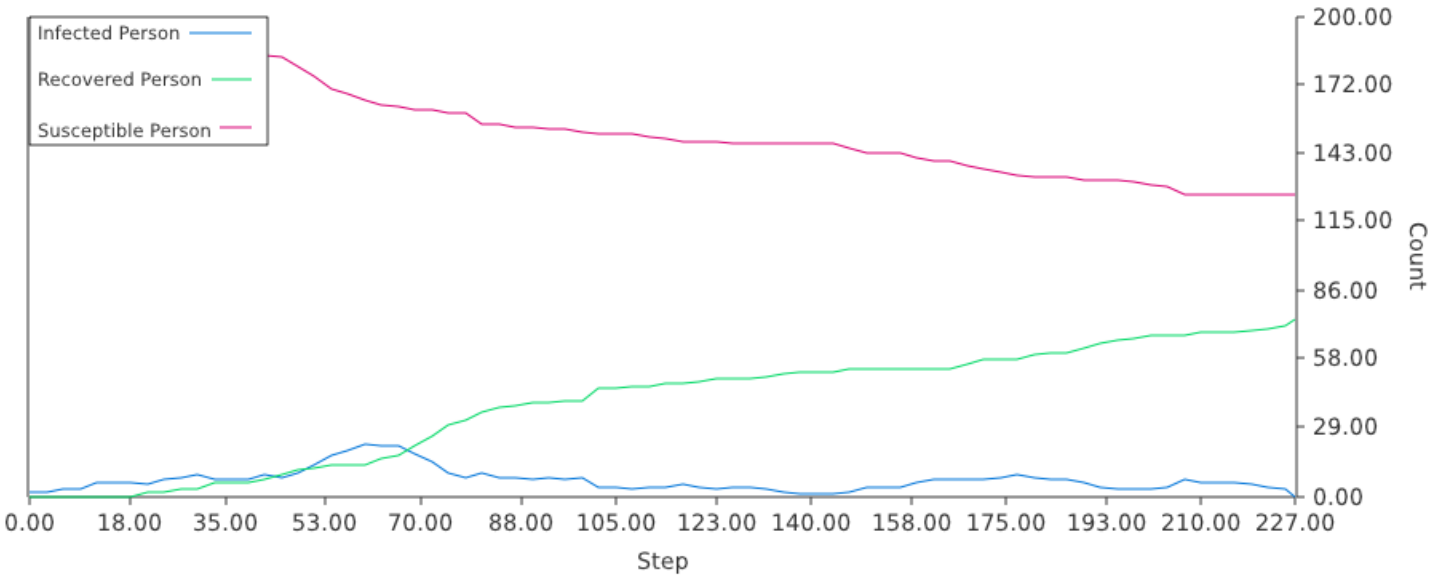
Steps completed: 228. Total agents (final): 200. Spawned: 146. Removed: 146. Interactions: 0. Events: 0. Rule executions: 47300.

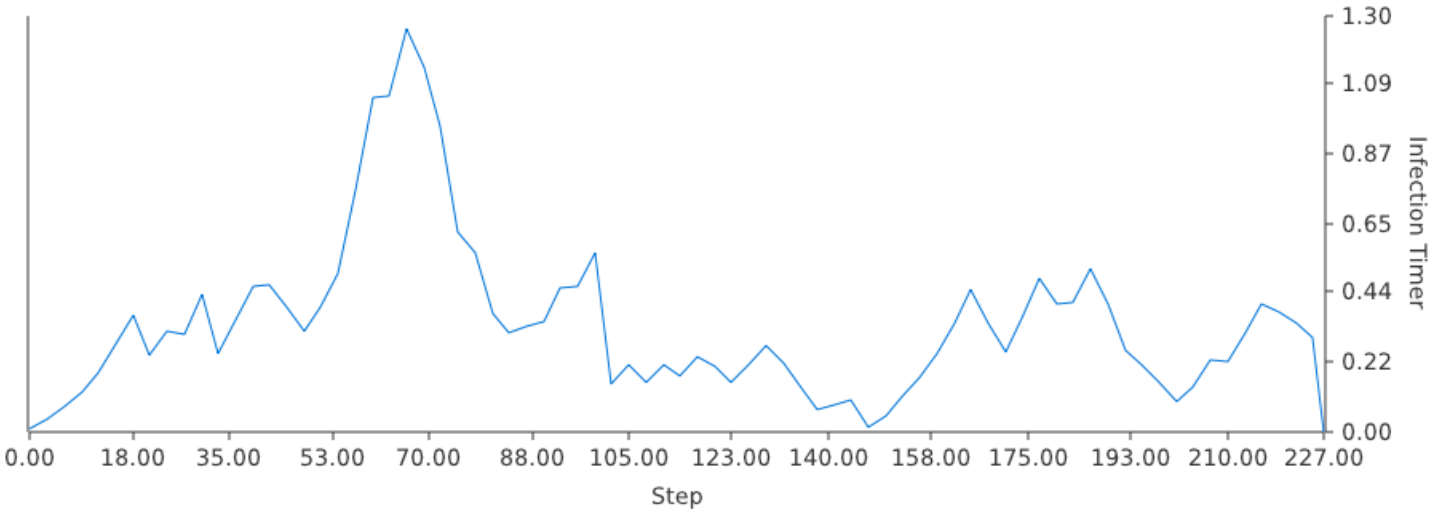
Archetype Susceptible Person — current 126, peak 198, min 126, spawned 0, removed 72.

Archetype Recovered Person — current 74, peak 74, min 2, spawned 74, removed 0.

Archetype Infected Person — current 0, peak 22, min 1, spawned 72, removed 74.

Population Over Time





Population & Property Dynamics: SIR Model in a Toroidal Grid

The simulation exhibits a classic outbreak phase followed by decline and a secondary ripple. Initially (steps 0–60), the infected population grew from 2 to a peak of 22 at step 60, while susceptible fell from 198 to 165. The average infection timer increased in lockstep, rising from 0.01 at step 0 to 1.045 at step 60, indicating accumulation of older cases.

After the peak, depletion of susceptibles drove a steady decline: infected dropped to a low of 1 by step 138, and susceptible stabilized around 147 between steps 132 and 144. Correspondingly, the average infection timer fell to 0.015 by step 147, reflecting rapid turnover and recovery of remaining cases.

A secondary wave emerged between steps 147 and 180 when infected climbed from 2 to 9 at step 177, fueled by the remaining susceptible pool (~145–133) and seeding from residual infections. The infection timer climbed again to 0.51 at step 186 before tapering off. By step 227, all infected had recovered, leaving 74 recovered and 126 susceptible.

Key findings

- Peak infection of 22 agents at step 60 coincided with susceptible dropping to 165.
- Average infection timer peaked at 1.045 at step 60, then fell to 0.015 by step 147.
- Primary decline reduced infected to 1 by step 138; susceptibles stabilized at ~147.
- Secondary wave peaked at 9 infected by step 177, despite reduced susceptible pool.

- Final state at step 227: 0 infected, 74 recovered, 126 susceptible.

Causal links

• Depletion of susceptibles (198→165 by step 60) → peak and subsequent decline of infected (22 at step 60 to 1 by step 138).

• Accumulation of infection age (timer ↑ to 1.045 at step 60) → signals imminent peak in active infections.

• Residual infected cases around step 147 → triggered secondary wave (infected ↑ from 2 at step 147 to 9 by step 177).

Emergent phenomena

- Bi-phasic outbreak with a secondary wave despite no influx of new susceptibles.
- Oscillatory coupling between average infection age and case counts, indicating generational turnover.

Spatial & Interaction Patterns

At initialization (Step 0), the 50 sampled susceptible agents are scattered across the 50×50 toroidal grid (x and y spanning 0–49), indicating no preexisting spatial clusters. Over 228 steps, 45 600 move_randomly executions and a total movement distance of ~49 684 units maintained strong mixing, preventing agents from settling into fixed regions.

Infections occurred when random walks temporarily brought susceptible and infected agents into proximity. The susceptible_gets_infected rule fired 72 times, producing a transient peak of 22 infected individuals (statistics peak) rather than stable hotspots. Snapshots at Step 113 reveal a brief local density at positions (14,35) and (14,34), but by Step 227 agents are again widely dispersed, showing that clusters dissolve quickly under uniform random motion.

Recovered agents (74 total) remain intermingled with susceptibles through the toroidal domain, with no evidence of spatial segregation or territory formation. Immunity did not create barriers—recovered individuals continued to move randomly, filling the same areas as susceptibles and any remaining infecteds until the epidemic ended.

Key findings

- Initial distribution is uniform across the toroidal grid with no seeded clusters.

• High frequency of random movement (45 600 moves) sustains thorough mixing, dissolving temporary gatherings.

• Infection events are spatially scattered, arising from fleeting local proximities rather than persistent hotspots.

• Transient clusters (e.g., two agents at (14,34–35) at Step 113) drive infections but disperse within a few steps.

- Recovered agents remain uniformly mixed; no segregation or territory formation occurs.

Causal links

• High rate of random movement executions (45 600) → strong mixing across grid → uniform spatial distribution preventing long-lived clusters

• Proximity-based infection rule executions (72) in transient local pockets → brief spikes in infected population (peak 22)

Emergent phenomena

• Spontaneous, short-lived infection waves emerge from random encounters without persistent spatial hotspots

- Global herd immunity arises through uniform mixing rather than spatial isolation

Emergent Behavior & Causal Relationships

The simulation exhibits a classic epidemic wave: after two initial infections at step 0, the number of infected agents grew nonlinearly to a peak of 22 at step 60. This rise was driven by random movement and proximity-based transmission, as seen in the jump from 6 infected at step 12 to 22 by step 60. Concurrently, the average infection timer climbed to a maximum of 1.26 by step 66, reflecting the accumulation of infection duration before recovery.

Recovery events began to outpace new infections after the peak: between steps 60 and 72, infected fell from 22 to 14 while recovered rose from 13 to 25. By step 227, all infections had resolved, leaving 74 recovered and 126 still susceptible. The decline in infections correlates with the shrinking susceptible pool, indicating a threshold-like suppression of transmission even though less than 50% had gained immunity.

Following the main wave, small secondary flare-ups appeared. Infections rose from 1 at step 144 to 4 by step 150, and again from 3 at step 198 to 7 at step 207. These transient rebounds reflect stochastic pockets of susceptibles encountering residual infected agents, producing oscillatory tails before eventual fade-out.

Overall, the interplay of movement, infection duration, and depletion of susceptibles produced an emergent epidemic curve with a single major wave, partial herd-immunity suppression, and minor secondary oscillations.

Key findings

- Peak of 22 simultaneous infections at step 60 following nonlinear growth from 2 initial cases.
- Average infection timer reached 1.26 at step 66, marking maximal infection duration before recovery.
- Total of 74 recoveries by step 227, with 126 susceptibles remaining—epidemic extinguished without full herd immunity.
- Major decline in infections occurred as recovered count rose from 13 at step 60 to 25 by step 72.
- Secondary flare-ups occurred between steps 144–150 and 198–207, producing small oscillatory infection spikes.

Causal links

- Random movement + proximity-based infection rule → rapid increase from 6 infected at step 12 to 22 by step 60.
- Increment of infection_timer beyond threshold → execution of infected_recovers rule → rise in recovered from 13 at step 60 to 25 at step 72.
- Accumulation of recovered agents → reduction in susceptible pool → decline in new infections after the peak.
- Residual infected agents encountering susceptible clusters → secondary flare-ups at steps 147–150 and 198–207.

Emergent phenomena

- Nonlinear epidemic wave with rapid exponential-like growth followed by decline.
- Threshold-driven suppression of transmission before full population immunity (partial herd immunity).
- Oscillatory tail behavior with secondary infection rebounds due to stochastic local interactions.

Conclusions

- The epidemic peaked at step 60 with 22 active infections; attack rate slowed as susceptibles fell from 198 to 165.
- Average infection timer rose to ~1.05–1.26 before the peak, indicating accumulation of case age prior to recovery surges.

• Primary wave declined to one case by step 138 as recoveries (25→52) outpaced new infections, demonstrating threshold-driven suppression before full population immunity.

• Two secondary flare-ups (steps 147–180, 198–207) arose from residual infected agents encountering pockets of susceptibles, peaking at 9 and 7 active cases.

• High-frequency random movement sustained uniform mixing, preventing long-lived spatial clusters and causing transmission to rely solely on fleeting local proximities.

Recommendations

• Test lower movement rates or heterogeneous mobility patterns to assess how spatial clustering alters peak magnitude and wave duration.

• Vary the infection_timer threshold or recovery rate to explore its impact on epidemic timing and secondary oscillations.

• Introduce temporary immunity waning or reinfection rules to investigate the potential for sustained endemic oscillations.

• Implement periodic introduction of new infected agents to study the resilience of herd-immunity suppression under continuous seeding.

• Incorporate spatially biased movement (e.g., hotspots or home ranges) to evaluate the emergence and persistence of infection clusters and their effect on overall dynamics.